

NICU Procedures

Pocket guide, procedure edition
Version 8, October 2026

Tags

- 8th NRP 8th edition
- 9th NRP 9th edition (effective 2026)
- Atlas MacDonald's Atlas of Procedures in Neonatology, 6th ed
- Baylor Baylor Guidelines for Acute Care of the Neonate, Ed. 34, where it differs

Sections

- Airway: ETT size, depth, tip, RSI
- Resus: NRP 8th vs 9th, SpO₂, emergency meds
- Lines: UAC, UVC, PICC
- Reference: x-ray targets, unit values, sources

Doses and depths are starting points. Confirm every tube and line clinically and on film, and follow unit protocol.

ETT depth

GA (wk)	Weight (g)	cm
<23 9th	<500	5.0–5.5
23–24	500–600	5.5
25–26	700–800	6.0
27–29	900–1000	6.5
30–32	1100–1400	7.0
33–34	1500–1800	7.5
35–37	1900–2400	8.0
38–40	2500–3100	8.5
41–43	3200–4200	9.0

9th values are measured **at the gum**. 8th used the same bands **at the lip**, confirmed in the Atlas. Atlas

Cross-checks

Nasal-tragus length + 1 cm. Weight (kg) + 6 cm tends to run deep, especially in ELBW infants.

RSI premedication

Typical order

1	Atropine	0.02 mg/kg IV or IM Atlas
2	Fentanyl	1–4 mcg/kg IV Atlas
3	Rocuronium	0.6–1.2 mg/kg IV Atlas Alternative: vecuronium 0.1 mg/kg IV

Fentanyl

Give it slowly: a fast push can cause chest wall rigidity. Naloxone or a paralytic reverses it. Atlas

There's no single standard regimen; follow your unit's order set.

Baylor practice Baylor

Atropine 0.02 mg/kg IV push. Fentanyl 0.5–2 mcg/kg IV over 3–5 min. Midazolam 0.1 mg/kg over 2 min if more sedation needed (Baylor's drug chart: avoid benzodiazepines <44 wk PMA). Vecuronium 0.1 mg/kg IV push, only after adequate sedation.

Pre-ductal SpO₂ targets

Time after birth	Target SpO ₂
1 min 8th only	60–65%
2 min	65–70%
3 min	70–75%
4 min	75–80%
5 min	80–85%
10 min	85–95%

Values from 2 min on are the same in both editions. 9th dropped the 1-minute row.

Starting FiO₂ 9th

- ≥35 wk: 21%
- 32–34 wk: 21–30%
- <32 wk: ≥30%

9th A reasonable approach is to adjust FiO₂ in 20–30% steps every 30 s until the target is reached.

ETT tip and confirmation

Tip position on film

Textbook Atlas	T1–T2 (also optimal in recent trials)
NNP program	T2–T3

Either way: between the clavicles and the carina, at least 0.5–1.0 cm above the carina (carina at about T4).

Head position moves the tube

Neck **flexion**: tube moves **down** (deeper).

Neck **extension**: tube moves **up**.

Confirm placement

- Rising heart rate
- Chest rise
- Symmetric breath sounds
- Exhaled CO₂ detector color change
- Chest x-ray

ETT size

ID (mm)	8th	9th
2.0	—	Optional for <800 g (22–25 wk), if available
2.5	<1000 g <28 wk	<800 g (22–25 wk) 800–1200 g (26–28 wk)
3.0	1000–2000 g 28–34 wk	1201–2200 g 29–34 wk
3.5	>2000 g >34 wk	>2200 g >34 wk

Where depth is measured

8th Tip to **upper lip**.

9th Tip to **front edge of the upper gum, midline**.

Supraglottic airway **9th**

Can be used as the primary device if face-mask ventilation or intubation is unsuccessful or unavailable (≥ 34 wk).

Baylor <1 kg or 22–27 wk: 2.0 or 2.5. 1–<2 kg or 28–34 wk: 3.0. ≥ 2 kg or 35+ wk: 3.5. Post-term or larger: 4.0.

Emergency access and meds

Emergency UVC

Low-lying: insert **2–4 cm below the skin**, just until blood returns. IO is the alternative.

Epinephrine 0.1 mg/mL

IV / IO	0.01–0.03 mg/kg = 0.1–0.3 mL/kg Suggested start 0.02 mg/kg = 0.2 mL/kg Flush 3 mL NS
ETT	0.05–0.1 mg/kg = 0.5–1 mL/kg Only while IV access is being obtained
Repeat	Every 3–5 min

Volume

NS **10 mL/kg** over 5–10 min, via UVC or IO.

NRP 8th vs 9th

	8th	9th
Cord clamping	Defer 30–60 s	Defer ≥ 60 s
Cord milking <28 wk	—	Not recommended (severe IVH risk)
Initial FiO₂	≥ 35 wk: 21% <35 wk: 21–30%	≥ 35 wk: 21% 32–34 wk: 21–30% <32 wk: $\geq 30\%$
SpO₂ table starts	1 min	2 min
PPV rate	40–60/min	30–60/min
Initial PIP	20–25 cm H ₂ O	≥ 32 wk: 25 (25–30) <32 wk: 20–25
PEEP	5	5
Initial steps	Includes "suction if needed"	Suction removed from the initial-steps box
Corrective steps	—	Start after 15–30 s of PPV
ETT	Depth to lip	Depth to gum; new size table

Resus doses by weight

kg	IV/IO epi 0.2 mL/kg	ETT epi 1 mL/kg	NS 10 mL/kg
0.5	0.1	0.5	5
0.75	0.15	0.75	7.5
1	0.2	1	10
1.5	0.3	1.5	15
2	0.4	2	20
2.5	0.5	2.5	25
3	0.6	3	30
3.5	0.7	3.5	35
4	0.8	4	40

All volumes in mL. Epinephrine is the 0.1 mg/mL concentration. IV/IO column is the suggested starting dose; the range is 0.1–0.3 mL/kg. ETT column is the top of the 0.5–1 mL/kg range.

UVC

Tip position

IVC–right atrial junction, just above the diaphragm.

On film: **T9–T10** per the textbook [Atlas](#); **T8–T9** per NNP program teaching. Follows the ductus venosus; sits to the right of the spine.

Insertion depth (cm), add stump length

Shukla	$[(3 \times \text{BW kg}) + 9] \div 2 + 1$ (half the high UAC + 1)
Revised	$[(3 \times \text{BW kg}) + 9] \div 2$ Dropping the +1 reduced over-insertion without more low lines
Surface	Umbilicus-to-nipple distance – 1 cm

Catheter [Atlas](#)

3.5 Fr <3.5 kg, 5 Fr >3.5 kg.

Confirm with AP and lateral films or ultrasound. Emergency UVC: see Resus section.

Baylor $((3 \times \text{kg}) + 9) \div 2$, target T9–T10. Catheter 3.5 Fr <1500 g, 3.5–5 Fr >1500 g. Remove by about 7 days.

Removing a UAC

Steps [Atlas](#)

1. Leave the umbilical tie loose around the stump.
2. Withdraw slowly until about 5 cm remain, then tighten the tie.
3. Stop the infusion.
4. Pull the rest at 1 cm per minute to allow vasospasm.
5. For bleeding, pinch the cord from the sides.

Reminders for every umbilical line

- Add the stump length to every formula.
- Confirm the tip on film before using the line.
- A formula is a starting point; the film decides.

X-ray tip targets

Device	Target
ETT	T1–T2 textbook; T2–T3 NNP program (carina $\approx$ T4)
UAC high	T6–T9 (ideal T7–T9)
UAC low	L3–L4
UVC	IVC–RA junction: T9–T10 textbook; T8–T9 NNP program
PICC, arm or scalp	SVC outside the heart, or SVC–RA junction
PICC, leg	Above L4–L5 or the iliac crest, not in the heart; get a lateral
PICC Baylor	Upper T3–T5; lower T8–T10

Unit policy decides.

Umbilical line depth by weight

BW kg	UAC Shukla	UAC Wright	UVC Shukla	UVC revised
0.5	10.5	9	6.3	5.3
0.75	11.3	10	6.6	5.6
1	12	11	7	6
1.25	12.8	12	7.4	6.4
1.5	13.5	13	7.8	6.8
2	15	—	8.5	7.5
2.5	16.5	—	9.3	8.3
3	18	—	10	9
3.5	19.5	—	10.8	9.8
4	21	—	11.5	10.5

Centimeters **before** adding stump length. Wright shown only to 1.5 kg because it was studied in VLBW infants. Rounded to 0.1 cm.

UVC revised = UVC Shukla minus 1 cm (it drops the +1, which tended to place lines too deep). Baylor uses the revised formula.

Doses, sizes, and sources

From MacDonald's Atlas [Atlas](#)

Rocuronium	0.6–1.2 mg/kg IV
Fentanyl	1–4 mcg/kg IV
UAC heparin	1–2 units/mL
UAC catheter	3.5 Fr <1200 g, 5 Fr >1200 g
UVC catheter	3.5 Fr <3.5 kg, 5 Fr >3.5 kg

Confirm doses with your unit's order set.

Sources

AAP NRP 9th ed launch communication (Oct 2025); CPS, NRP 9th ed changes for busy clinicians (2026); In[Neo]Sight ETT calculator (NRP 9th tables); PubMed 40739063 (ETT tip); NCCC umbilical line guidelines, mombaby.org (2022); Medscape UAC (2024); PubMed 23503982 (revised UVC); NCT02939690 (UN–1); Iran J Pediatr 2024, citing NANN (PICC); Pediatr Neonatol 2023, POCUS PICC; neonatal atropine RCT; VanLooy NICU premedication algorithm; PMC8228479 (emergency UVC); NRP textbook via tomwademd.net (volume); MacDonald's Atlas of Procedures in Neonatology, 6th ed (ch. 31, 32, 38); program NNP Brain reference.

UAC

Tip position

High (preferred): T6–T9, above the celiac axis; ideal T7–T9.

Low: L3–L4, above the aortic bifurcation (L4–L5). Low lines displace more easily and are not preferred.

Insertion depth (cm), add stump length

Shukla	$(3 \times \text{BW kg}) + 9$
Wright	$(4 \times \text{BW kg}) + 7$ Studied in VLBW infants

Precomputed depths are on the Lines depth table page.

Catheter and flush [Atlas](#)

3.5 Fr <1200 g, 5 Fr >1200 g. Heparin 1–2 units/mL in 0.45–0.9 NS.

Confirm tip position on film before use.

[Baylor](#) Shukla for all weights. Catheter 3.5 Fr <1500 g, 3.5–5 Fr >1500 g. Remove by about 5 days.

PICC

Measure [Atlas](#)

Arm	Arm abducted 90°, head turned toward the insertion side. Site → along the vein → suprasternal notch → 3rd right intercostal space.
Scalp	Site → along the vein → jugular → right sternoclavicular joint → 3rd right intercostal space.
Leg	Site → along the vein → xiphoid.

Tip position [Atlas](#)

Arm or scalp: SVC outside the heart, or at the SVC–RA junction.

Leg: above L4–L5 or the iliac crest, not in the heart.

Film tips

You can insert with the arm in any position. Before the x-ray, return the arm to the position the infant usually lies in.

A leg line can divert into the ascending lumbar vein and still look correct on an AP film. Get a lateral.

[Baylor](#) Upper body SVC–RA junction, T3–T5. Lower body thoracic IVC, T8–T10. Not in the RA; consider replacing a lower-body line at L2–L3.